Lecture 15 recap

In Lecture 15:

- Introduced eigenvalues and eigenfunctions for $-\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ with periodic BCs
  - They are $\cos(k_n\, x)$ and $\sin(k_n\, x)$ for $k_n = n\,\pi/L$ $(n = 1, 2, \ldots)$
  - and the constant function (we chose $1/2$)
- Showed they are orthogonal with respect to inner product $\langle f, g\rangle = \displaystyle\int_{-L}^{L} f\, g\, \mathrm{d}x$

Lecture 16

In Lecture 16, we will:

1. Introduce the property of self-adjointness
2. Define Fourier series
3. Introduce the Euler–Fourier formulae
4. Calculate Fourier series for example functions

Generalising real symmetric matrices

Self-adjoint linear transformations

Let T be a linear transformation. T is self-adjoint if

$$\langle \mathsf{T} f, g \rangle = \langle f, \mathsf{T} g \rangle$$

Consider the eigenvalue problem

$$\mathsf{T} y = \lambda y$$

Then T has real-valued eigenvalues λ.

Example:

$\mathsf{T} = -\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}$ with periodic BCs is self adjoint under the inner product from last lecture

Fourier series

A *Fourier series* on $x \in (-L, L]$ has the form

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x) \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

for some set of coefficients a_n and b_n

- Each term in the Fourier series is called a *mode*
- The coefficients a_n and b_n are also known as the *amplitudes* of the modes
- k_n are *wavenumbers*
- If independent variable is time we use *angular frequency*, ω_n, instead of k_n

Fourier series

- The Fourier series is $2L$-periodic

Fourier series: Coefficients

Suppose that the Fourier series converges for all $x \in (-L, L]$ and that

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x)$$

The Fourier coefficients are given by the Euler–Fourier formulae

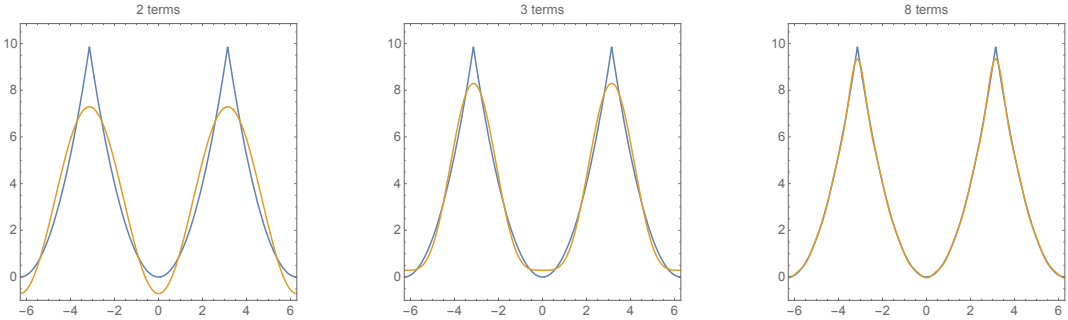
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(k_n x)\,\mathrm{d}x \quad \text{and} \quad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(k_n x)\,\mathrm{d}x$$

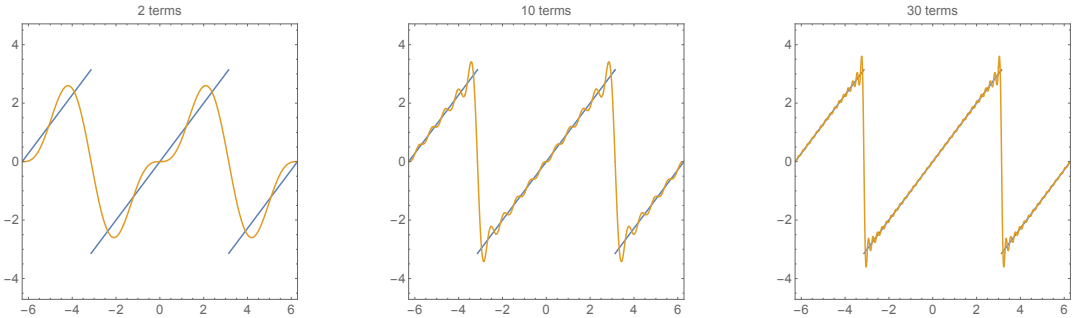
Exercise 4.3: Find the Fourier series of $f(x) = x^2$ where $x \in (-\pi, \pi]$

Exercise 4.4: Find the Fourier series of $f(x) = x$ where $x \in (-\pi, \pi]$

Fourier series of x^2

The Fourier series of $f(x) = x^2$ on $x \in (-\pi, \pi]$ is converging to its *periodic extension*.



Fourier series of x

The Fourier series of $f(x) = x$ on $x \in (-\pi, \pi]$ is converging to its *periodic extension*.